AI Capability Boundary Reference

AI Literacy for the Modern Workforce

KNOW THE BOUNDARY
— before you delegate

Know what the tool does well and where it systematically fails before you delegate.
Organized by task type, not technical architecture. Each entry connects to Module 3's mechanical framework.

WHERE AI IS RELIABLE

Tasks where models produce useful output under well-specified prompts. Reliability here means errors are infrequent rather than absent.

TASK TYPE	WHY IT WORKS	NOTES
Drafting and composition	Dense training patterns for written formats (emails, memos, reports). Well-specified prompts produce well-structured output.	Quality scales with Description quality. Specify product, process, and performance.
Summarizing and compression	The model excels at identifying high-probability patterns, which is what compression requires.	Verify completeness; the model may drop low-frequency details.
Reformatting and restructuring	Deterministic tasks with clear output specifications. Token-level manipulation within well-defined patterns.	Format + audience + length spec → reliable output.
Brainstorming and ideation	High-temperature sampling across the probability distribution produces diverse options by design.	Useful for breadth. Evaluate each option independently; plausible ≠ good.
Translation (major languages)	Dense training representation for high-resource language pairs (English ↔ Spanish, French, German, Mandarin).	Reliability drops for lower-resource languages due to tokenization asymmetry. See page 2.
Code generation (common patterns)	Well-documented languages and frameworks are heavily represented in training data.	Edge cases, niche libraries, and novel architectures are less reliable. Test all output.
Pattern recognition in text	Categorization, sentiment analysis, entity extraction: tasks where the model identifies structure.	Reliable for well-defined categories. Less reliable for subjective or culturally specific judgments.

Framework: Anthropic 4D AI Fluency Framework, Delegation (Platform Awareness) and Discernment competencies
(Dakan, Feller, and Anthropic, 2025). Mechanical explanations from Module 3.

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WHERE AI SYSTEMATICALLY FAILS

Three categories of unreliable tasks from Module 3. Each traces to a mechanical root cause.

01 BOUNDARY TASKS

ROOT CAUSE **Tokenization**

Token boundaries misalign with the meaningful structure of the input. Errors concentrate where tokens and meaning diverge.

FAILS AT	WHY	INSTEAD
Multi-digit arithmetic	Numbers split into arbitrary token chunks; "127" may become ["12", "7"] or ["1", "27"]	Ask the model to write code that performs the calculation
Character counting and spelling	The model sees tokens rather than letters; "strawberry" is not processed letter by letter	Use programmatic tools for character-level tasks
Non-English text processing	Lower-resource languages fragment 2–3× more aggressively, consuming context window and increasing error surface	Verify non-English output more carefully; expect quality gaps

02 SPECIFICITY TASKS

ROOT CAUSE **NTP × Knowledge**

The model generates statistically probable tokens. When a task requires specific facts the training data represents thinly, it produces plausible fabrications.

FAILS AT	WHY	INSTEAD
Specific citations and references	Citation-shaped text is common in training data, so the model generates format-perfect references that may not exist	Verify every citation against the original source
Precise statistics and figures	Numbers generated to satisfy the pattern, not retrieved from a database	Check every specific number against primary data
Legal / regulatory references	Regulation numbers, case names, and statutory language are specificity-zone content	Cross-reference against official legal sources
Named quotes and attributions	Quote-shaped text attributed to specific people is generated by probability, not recall	Confirm the quote exists in the attributed source

03 VOLUME TASKS

ROOT CAUSE **Context window**

The model has a finite processing window. Information outside is not processed. Information in the middle receives less attention.

FAILS AT	WHY	INSTEAD
Long document analysis (>20 pages)	Middle sections fall into the attention valley: processed but underweighted	Break long documents into sections; process separately
Sustained instruction fidelity	Over many turns, earlier instructions lose influence as context fills	Restate critical instructions periodically; compare to original spec
Multi-document synthesis	Multiple documents consume context rapidly; later documents may be underprocessed	Process documents individually, then synthesize yourself

THE CORE DISTINCTION

The model *generates* text rather than *retrieving* facts. Fluent output and accurate output are independent properties.

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